DSCI-560: Data Science Practicum Final Project Proposal

Domain-Specific AI Enhanced Group Chat System

Instructor: Young H. Cho, Ph.D.

1) The Goal: Real Solution to a Real-World Problem

2) For Monday Lecture: In-person Group Meeting
   a) Set aside time to meet with your teammates before Monday's lecture to discuss at least **THREE** specific domains/topics of interest with known problems that a group of people within that domain face.
   b) Brainstorm about ways that a domain expert can help that group of people.
      i) What information/data do they need?
      ii) Where can one find that information/data?
      iii) What services can a domain expert provide?
   c) Think about creative ways that a Group Chat App can be used to help that group of people
   d) Think about creative ways that an AI Chatbot with Embeddings can be used to play the role of the domain expert within that chat interface
   e) During Tuesday lecture, I would like to go to each team to talk about what your team discussed and attempt to guide the team to select the best ideas to pursue

3) For Thusrday Lecture: Final Project Proposal Presentations
   a) Prepare about 5 PowerPoint Slides to discuss your selected Final Project Idea
   b) Include at least **THREE** references that indicate that the real-world problem exists
   c) During Thursday lecture, each team will be presenting the idea.
   d) After the presentation, discuss the final project with the team to finalize the selection and complete and submit the Proposal Presentation to Coursistant website by Friday night.

4) Minimum Requirements for the Project
   a) There MUST be real-world services (human or digital) that attempt to solve the problem in that domain. The existing company should be profitable.
   b) Interview at least 9 real-life human beings in-person, Zoom, or over the phone (not a survey) to discuss the problem and how much they are willing to pay for a solution. The recordings of the interviews should be included in the final demo video/presentation. You must get their permission.
   c) There must be both Web interface and App.
   d) The chat interface should allow you to upload files or pictures.
   e) The AI solution must use embeddings.
   f) The AI agent must be able to display pictures.